Course: BTech **Semester:** 7

Prerequisite: Software Engineering Basics, Basics of Java Programming

Course Objective: To study pioneer of Software Development Life Cycle, Development models and Agile Software development. To study fundamental concepts in software testing, including software testing objectives, process, criteria, strategies, and methods. To discuss various software testing issues and solutions in software unit test; integration, regression, and system testing. To learn the process of improve the quality of software work products. To gain the techniques and skills on how to use modern software testing tools to support software testing projects. To expose Software Process Improvement and Reengineering

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Software Testing**Testing, Verification and Validation, Test Strategies for Conventional and Object-OrientedSoftware, Unit Testing, Integration Testing, Validation Testing, AlphaandBetaTesting,SystemTesting ,Recovery Testing, Security Testing, Stress Testing,PerformanceTesting,MetricsforSourceCode,MetricsforTesting,Debugging Process, Debugging Strategies.	15	8
2	**Testing Techniques**Software Testing Fundamentals, Black Box and White Box Testing, Basis Path Testing,Flow Graph Notation, Independent Program Paths, Graph Matrices, ControlStructureTesting,ConditionTesting, Data Flow Testing, Loop Testing, GraphBasedTestingMethods,EquivalencePartitioning,Boundary Value Analysis.	15	7
3	**Object Oriented Testing Methods**Applicability of Conventional Test Case Design Methods, Issues in Object Oriented Testing, Fault-Based Testing, Scenario-Based Testing, Random TestingandPartitionTestingfor Classes, InterClass Test Case Design	20	7
4	**Testing Process and Specialized Systems Testing**TestPlanDevelopment, Requirement Phase,DesignPhase and Program Phase Testing, Testing Client/Server Systems, Testing Web based Systems, Testing Offthe-Shelf Software, Testing in Multiplatform Environment, Testing for Real Time Systems, Testing Security.	15	7
5	**Software Quality Assurance Concepts and Standards** QualityConcepts,QualityControl,QualityAssurance,SQAActivities,SoftwareReviews,FormalTechnicalReviews,ReviewGuidelines,SoftwareReliability,SoftwareSafety,QualityAssuranceStandards,ISO9000,ISO9001:2000,ISO9126 QualityFactors, CMM, TQM, Six Sigma, SPICE, Software Quality Assurance Metrics	20	8
6	**Risk Management and Change Management**Software Risks, Risk Identification, Risk Projection, Risk Refinement, The RMMM Plan, Software Configuration Management, Baselines, SoftwareConfigurationItems, SCM Process: Version Control, Change Control, ConfigurationAudit,ConfigurationManagementforWeb Engineering.	15	8

Reference Books

1.	**Software Engineering** By R. Pressmen | 6th
2.	**Software testing** By Yogesh Singh | Cambridge University Press, 2012
3.	**Effective Methods for Software Testing** By William Perry

Course Outcome

After Learning the Course the students shall be able to:
1. Prepare SRS (Software Requirement Specification) document and SPMP (Software ProjectManagement Plan) document. 2. Apply the concept of Functional Oriented and Object Oriented Approach for Software Design. 3. Recognize how to ensure the quality of software product, different quality standards and softwarereview techniques. 4. Apply various testing techniques and test plan in. 5. Able to understand modern Agile Development and Service Oriented Architecture Concept of Industry.